Singular Limits for Three-Dimensional Global Minimizers of Ginzburg–Landau-Type Functionals

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Conference of GMT and PDE, Enshi Hubei

Aug. 20 2026

Outline

1. Background and setting
2. Topological energy and network cost
3. Main results
4. Key estimates and proof strategy
5. Summary

The functional and its singular limit

Domain and target: Let $\Omega \subset \mathbb{R}^3$ be bounded, and let $\mathcal{N} \subset \mathbb{R}^m$ be a smooth, compact, connected manifold isometrically embedded in $\mathbb{R}^m$.

For $u \in H^1(\Omega, \mathbb{R}^m)$ and $0 < \varepsilon < 1$, set

$$E_\varepsilon(u, \Omega) := \int_\Omega e_\varepsilon(u)\,\mathrm{d}x, \quad e_\varepsilon(u) := \frac{1}{2}|\nabla u|^2 + \frac{1}{\varepsilon^2} f(u), \tag{GL$_\varepsilon$}$$

where $f \geq 0$ and $f^{-1}(0) = \mathcal{N}$.

Critical points: If $f \in C^1$, then

$$\Delta u = \frac{1}{\varepsilon^2} Df(u).$$

Singular limit: As $\varepsilon \downarrow 0$, the potential drives u toward $\mathcal{N}$ away from a defect set; topology may force that set to persist.

Why the logarithmic scale?

For the classical complex model,

$$\mathcal{N} = \mathbb{S}^1, \quad f(u) = \frac{1}{4}(|u|^2 - 1)^2.$$

2D and 3D

- In dimension two, the minimum energy in degree $d \neq 0$ is

$$\pi|d|\,|\log\varepsilon| + O(1)$$

  [Bethuel–Brezis–Hélein, 1994; Struwe, 1994].
- In dimension three, the transverse vortex profile may extend along a curve; its leading-order energy is proportional to its length.

Natural scale in three dimensions:

$$\limsup_{\varepsilon\downarrow 0} \frac{E_\varepsilon(u_\varepsilon, \Omega)}{|\log\varepsilon|} < +\infty, \quad \dim S_* = 1.$$

Minimizers vs. Solutions

$\mathcal{N} = \mathbb{S}^1$, $n \geq 3$: logarithmic energy concentrates in codimension two.

Global minimizers [Lin–Rivière, 1999]

- Uniform convergence off the singular set.
- the limit is an integral rectifiable, "area-minimizing" $(n-2)$-varifold.

General solutions

- For $n = 3$: [Lin–Rivière, 2001];
- for all $n \geq 3$: [Bethuel–Brezis–Orlandi, 2001].
- The limit is stationary, possibly non-integral [Pigati–Stern, 2023].
- The limit can be unstable, geodesic [Colinet–Jerrard–Sternberg, 2025].

Two energy regimes

- As a one-dimensional Riemannian manifold, $\mathbb{S}^1$ is intrinsically flat.
- A general $\mathcal{N}$ may have nonzero intrinsic curvature; in dimension three, point singularities and line defects may coexist.

Bounded energy

$$\sup_{0<\varepsilon<1} E_\varepsilon(u_\varepsilon) < +\infty.$$

For global minimizers, the limit is a minimizing harmonic map u_*, with

$$\dim_{\mathcal{H}} \operatorname{sing}(u_*) \leq n-3$$

(points when $n=3$).
[Lin–Wang, 1999; Majumdar–Zarnescu, 2010; Contreras–Lamy–Rodiac, 2018; Contreras–Lamy, 2022].

Logarithmic energy

$$\sup_{0<\varepsilon<1} \frac{E_\varepsilon(u_\varepsilon)}{|\log\varepsilon|} < +\infty.$$

$\pi_1(\mathcal{N}) \neq 0$ supports codimension-two line defects, labeled by free homotopy classes.
For $\mathbb{S}^1$, $\pi_1(\mathbb{S}^1) \simeq \mathbb{Z}$, so these labels reduce to integer degrees.

If $\pi_1(\mathcal{N})$ is non-Abelian, there is no single integer degree: charge addition can be multi-valued.

Liquid-crystal examples

Landau–de Gennes (LdG) theory uses a tensor order parameter Q; its bulk energy wells form the vacuum manifold $\mathcal{N}$.

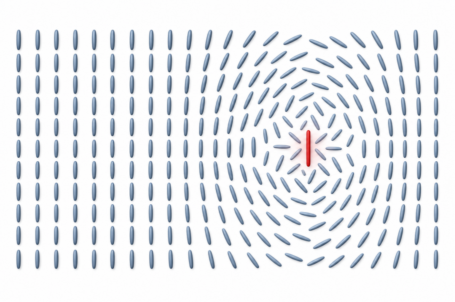
- **Uniaxial wells:** $n \sim -n$, hence

$$\mathcal{N} \simeq \mathbb{RP}^2.$$

- **Biaxial wells:**

$$\mathcal{N} \simeq \mathbb{S}^3 / Q_8.$$

- $\pi_1(\mathcal{N}) \neq 0$ produces non-trivial line disclinations.



ordered nematic line disclination

What was known before

- **Two-dimensional vortices**:
  - Convergence and defect-core analysis for uniaxial Landau–de Gennes [Canevari, 2015];
  - For general $\mathcal{N}$, compactness away from finitely many vortex points and renormalized-energy selection of their locations [Monteil–Rodiac–Van Schaftingen, 2021, 2022].
- **Three-dimensional interior networks**: Results for $\mathbb{RP}^2$ [Canevari, 2017] and $\mathbb{S}^3/Q_8$ [Wang–Zhang, 2024].
- **All dimensions**: Γ-convergence when $\pi_1(\mathcal{N})$ is Abelian [Canevari–Orlandi, 2021].

Three open tasks in dimension three.

- Prove compactness at logarithmic scale and identify a finite weighted network of straight segments.
- Determine the boundary endpoints and prove uniform estimates up to $\partial\Omega$.
- Establish weighted-length minimality of the labeled network.

Assumptions in dimension three

Domain: $\Omega \subset \mathbb{R}^3$ is bounded and $C^{2,1}$, and $\overline{\Omega}$ is strongly convex at every boundary point; hence $\partial\Omega$ is simply connected.

Vacuum manifold:

- $f \in C^\infty(\mathbb{R}^m, [0, +\infty))$;
- $\mathcal{N} = f^{-1}(0)$ is a compact, connected, smooth manifold without boundary;
- $\#\pi_1(\mathcal{N}) < +\infty$.

Coercivity and non-degeneracy:

$$\liminf_{|y|\to\infty} f(y) > 0, \quad y \cdot Df(y) \geq 0,$$

for $|y|$ large, and

$$D^2 f(x)[v, v] > 0 \quad \text{for } 0 \neq v \perp T_x\mathcal{N}.$$

Why require a finite fundamental group?

Finite $\pi_1(\mathcal{N})$ yields the uniform L^1 gradient estimate used to prove $W^{1,q}$ bounds. Without finiteness this mechanism may fail. For $\mathcal{N} = \mathbb{S}^1$, even

$$E_\varepsilon = o(|\log \varepsilon|)$$

need not imply local $W^{1,1}$ bounds or L^1_{loc} compactness [Bethuel–Brezis–Orlandi, 2001, Remark III.4].

Boundary data and prescribed charges

Boundary charges and limiting trace

Let $\mathcal{A} = \{a_1, \ldots, a_k\} \subset \partial\Omega$. The maps $g_\varepsilon \in C^2(\partial\Omega, \mathbb{R}^m)$ regularize these charges at scale ε, with

$$g_\varepsilon \to g \quad \text{in } C^2_{\mathrm{loc}}(\partial\Omega \setminus \mathcal{A}), \quad g \in C^2(\partial\Omega \setminus \mathcal{A}, \mathcal{N}).$$

Suitable boundary data

- **Scale-invariant control:**
$$|D^j_{\partial\Omega} g_\varepsilon(x)| \le M_0 \big(\max\{\mathrm{dist}(x, \mathcal{A}), \varepsilon\}\big)^{-j}, \quad j = 0, 1, 2;$$
- **Exact vacuum trace:** $g_\varepsilon = g \in \mathcal{N}$ away from the radius-ρ cores whenever $\varepsilon < \rho$;
- Non-trivial charge: every loop $g_\varepsilon|_{\partial\Omega \cap \partial B_\rho(a_i)}$ is non-trivial in $\mathcal{N}$;
- **Controlled angular core:** After flattening and rescaling, each core has a uniformly controlled angular profile.

Regularizing each boundary core

After flattening near a_i and rescaling by ε, the angular-core hypothesis prescribes

$$g_\varepsilon(x) = g_\varepsilon^0\left(\frac{\Phi_0(R(x-a_i))}{\varepsilon}\right), \quad g_\varepsilon^0(y) = h_\varepsilon\left(\frac{y'}{|y'|}\right)\chi(|y'|).$$

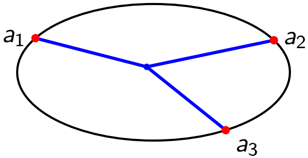
- $\Phi_0(0) = 0$, $D\Phi_0(0) = \mathrm{Id}$, while $h_\varepsilon : \mathbb{S}^1 \to \mathcal{N}$ is uniformly C^2.
- This is the hypothesis of [Lin–Rivière, 1999, p. 243(iii)]; it makes each g_ε a smooth ε-core regularization.
- In contrast with the p-harmonic model [Bulanyi–Van Schaftingen–Van Vaerenbergh, 2025], the potential and the ambient-valued core create additional boundary errors.

The angular structure controls the boundary stress–energy error down to scale ε.

The variational problem

u_ε is a global minimizer with trace g_ε:

$$u_\varepsilon \in \operatorname{argmin}\left\{E_\varepsilon(v,\Omega) : \operatorname{tr} v = g_\varepsilon\right\}.$$



Normalized energy measure:

$$\mu_\varepsilon := \frac{e_\varepsilon(u_\varepsilon)}{|\log\varepsilon|}\,\mathrm{d}x.$$

Weak-* limits of μ_ε record the leading-order energy concentration.

The problem is to identify the support of μ_*, the limit map u_*, and the network selected by global minimality.

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Free homotopy and line defects

Definition (Free homotopy)

Two continuous loops $\ell_1, \ell_2 \in C^0(\mathbb{S}^1, \mathcal{N})$ satisfy $\ell_1 \sim \ell_2$ if there exists

$$G \in C^0([0,1] \times \mathbb{S}^1, \mathcal{N}), \quad G(0,\cdot) = \ell_1, \quad G(1,\cdot) = \ell_2.$$

The quotient $[\mathbb{S}^1, \mathcal{N}]$ is the set of free homotopy classes.

Algebraic description: Choosing a base point gives

$$[\mathbb{S}^1, \mathcal{N}] \longleftrightarrow \{\text{conjugacy classes of } \pi_1(\mathcal{N})\}.$$

Label of a line defect: For a disk D transverse to the defect, assume $u|_{\partial D} \in C^0(\partial D, \mathcal{N})$; then define

$$\sigma = [u|_{\partial D}]_{\mathcal{N}}.$$

The class σ is unchanged as D slides through a regular tube that avoids the other defects.

The cost of a homotopy class

Least loop energy in a fixed class:

$$E_{\min}(\sigma) := \inf\left\{\frac{1}{2}\int_{\mathbb{S}^1} |u'|^2 \,\mathrm{d}\mathcal{H}^1 : u \in H^1(\mathbb{S}^1, \mathcal{N}),\ [u]_{\mathcal{N}} = \sigma\right\}.$$

Definition (Effective cost after optimal splitting)

$$|\sigma|_* := \inf\left\{\sum_{i=1}^{k} E_{\min}(\sigma_i) : \begin{array}{l} k \in \mathbb{N},\ \sigma_i \in [\mathbb{S}^1, \mathcal{N}], \\ \sigma \in \sigma_1 + \cdots + \sigma_k \end{array}\right\}.$$

The sum is understood in the iterated multi-valued sense.

The assumption $\#\pi_1(\mathcal{N}) < \infty$ gives a positive energy gap:

$$|\sigma|_* = 0 \iff \sigma = 0, \quad |\sigma|_* \geq c_0 > 0 \quad (\sigma \neq 0),$$

while the definition makes $|\cdot|_*$ subadditive under charge merging.

Junction topology versus geometry

For free homotopy classes, addition is induced by multiplication of conjugacy classes:

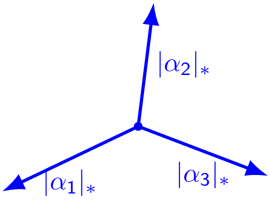
$$\alpha + \beta = \{\text{conjugacy classes of } ab : a \in \alpha,\ b \in \beta\}.$$

Topological compatibility: For incident labels $\alpha_1, \dots, \alpha_\ell$,

$$0 \in \alpha_1 + \cdots + \alpha_\ell.$$

Force balance: After stationarity is established, the tangent directions satisfy

$$\sum_{j=1}^{\ell} |\alpha_j|_* v_j = 0.$$



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Normalized energy measures

Energy bounds: Global minimality and the boundary construction give

$$E_\varepsilon(u_\varepsilon,\Omega) \le C(|\log\varepsilon|+1), \quad \|u_\varepsilon\|_{L^\infty(\Omega)} \le C.$$

Normalize the energy density: Set

$$\mu_\varepsilon := \frac{e_\varepsilon(u_\varepsilon)}{|\log\varepsilon|}\,\mathrm{d}x, \quad \mu_\varepsilon(\overline{\Omega}) = \frac{E_\varepsilon(u_\varepsilon,\Omega)}{|\log\varepsilon|} \le C\left(1+\frac{1}{|\log\varepsilon|}\right) \le C'$$

for $0<\varepsilon\le\frac{1}{2}$. Thus $\{\mu_\varepsilon\}$ has uniformly bounded total mass along any vanishing sequence.

Weak-* compactness: Along a sequence $\varepsilon_i \downarrow 0$, for some $\mu_* \in \mathcal{M}^+(\overline{\Omega})$,

$$\mu_{\varepsilon_i} \overset{*}{\rightharpoonup} \mu_* \quad \text{in } (C^0(\overline{\Omega}))'.$$

The concentration set is the support of the limiting measure:

$$S_* := \operatorname{supp}\mu_*.$$

Compactness away from the network

Theorem (Canevari–Fu–W., 2026)

The interior concentration set is 1-rectifiable and has finite length:

$$\mathcal{H}^1(S_* \cap \Omega) < +\infty.$$

Moreover, there exists

$$u_* \in H^1(\Omega \setminus S_*, \mathcal{N})$$

which locally minimizes the Dirichlet energy, and

$$u_{\varepsilon_i} \to u_* \quad \textit{strongly in } H^1_{\mathrm{loc}}(\Omega \setminus S_*, \mathbb{R}^m).$$

Residual point singularities: The set $S_0 := \mathrm{sing}(u_*)$ is locally finite in $\Omega \setminus S_*$. On $\Omega \setminus (S_* \cup S_0)$,

$$u_* \in C^\infty(\Omega \setminus (S_* \cup S_0), \mathcal{N}), \quad u_{\varepsilon_i} \to u_* \quad \text{in } C^0_{\mathrm{loc}}.$$

The limiting network

Theorem (Canevari–Fu–W., 2026)

There is a finite family of closed line segments $L_j \subset \overline{\Omega}$, with pairwise disjoint relative interiors, such that

$$S_* = \bigcup_{j \in J} L_j.$$

For $x \in \operatorname{Int} L_j$, choose a transverse disk D with $D \cap S_* = \{x\}$ and $\partial D \cap S_0 = \varnothing$. Define

$$\alpha_j := [u_*|_{\partial D}]_{\mathcal{N}} \neq 0, \quad \sigma_j := |\alpha_j|_* = \left|[u_*|_{\partial D}]_{\mathcal{N}}\right|_* > 0.$$

Then the limiting measure is the weighted length measure

$$\mu_* = \sum_{j \in J} \sigma_j \, \mathcal{H}^1 \llcorner L_j.$$

L_j: geometry $\quad$ α_j: meridian class $\quad$ σ_j: energy per unit length

Density quantization and force balance

The meridian class α_j, and hence the line tension σ_j, is independent of the choices of $x \in \operatorname{Int} L_j$ and the admissible transverse disk D.

Pointwise quantization: The one-dimensional density is exactly the transverse topological cost:

$$\Theta^1(\mu_*, x) := \lim_{r \to 0^+} \frac{\mu_*(\overline{B}_r(x))}{2r} = \sigma_j = \left|[u_*|_{\partial D}]_{\mathcal{N}}\right|_*.$$

Junction law: At an interior junction x, let v_j be the outward unit tangent of the j-th incident segment. Stationarity then yields

$$\sum_{j \in J_x} \sigma_j v_j = 0.$$

Boundary anchoring

Strong convexity and the prescribed charges give

$$S_* \cap \partial\Omega = \mathcal{A}, \quad S_* \subset \mathrm{Conv}(\mathcal{A}).$$

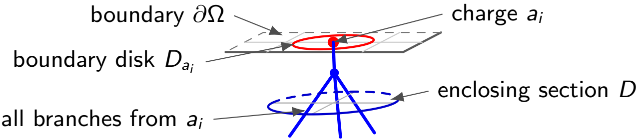
Thus there is no concentration on $\partial\Omega \setminus \mathcal{A}$; endpoints lie in $\mathcal{A}$ or at interior branching points.

Matching the boundary charge: Choose $D_{a_i} \subset \partial\Omega$ around a_i and an interior section $\overline{D} \subset \Omega$ enclosing all branches issuing from a_i, with

$$\partial D \cap \big(\mathrm{Conv}(\mathcal{A}) \cup S_0\big) = \varnothing.$$

A regular annulus joins ∂D_{a_i} to ∂D; hence

$$[g|_{\partial D_{a_i}}]_{\mathcal{N}} = [u_*|_{\partial D}]_{\mathcal{N}}.$$



Uniform estimates up to the boundary

Theorem (Canevari–Fu–W., 2026)

For any $q \in (1,2)$, the following estimates hold uniformly for $0 < \varepsilon < 1$:

$$\|\nabla u_\varepsilon\|_{L^q(\Omega)} \le C_q, \quad \frac{1}{\varepsilon^2}\int_\Omega f(u_\varepsilon)\,\mathrm{d}x \le C.$$

Both estimates hold on all of Ω, including neighborhoods of $\partial\Omega$.

- **Interior precedents:** Complex GL [Bourgain–Brezis–Mironescu, 2004; Bethuel–Orlandi–Smets, 2005] and Landau–de Gennes [Fu–Wang–W., 2026].
- **Boundary statement:** The $W^{1,q}$ and potential bounds are uniform on all of Ω, including neighborhoods of $\partial\Omega$.
- **Proof and sharpness:** Boundary partial regularity replaces the missing p-harmonic control of the potential. The endpoint $q = 2$ fails in general, while the potential energy remains on the optimal $O(1)$ scale.

The admissible class $\mathcal{C}(U, h)$

Competitor data: Let $U \subset \mathbb{R}^3$ be bounded Lipschitz, $h \in H^{\frac{1}{2}}(\partial U, \mathcal{N})$, and I be finite. Write

$$\Gamma = \{(K_i, \beta_i)\}_{i \in I}, \quad S_\Gamma := \bigcup_{i \in I} K_i.$$

Here $K_i \subset \overline{U}$ is a closed segment and $\beta_i \in [\mathbb{S}^1, \mathcal{N}]$.

Geometry:

- finitely many positive-length closed segments with pairwise disjoint relative interiors;
- every endpoint lies on ∂U or is an interior branching point;
- $S_\Gamma \cap \partial U$ is finite.

Realization by a map:
There exist a locally finite set $F \subset U \setminus S_\Gamma$ and a map

$$v \in H^1_{\mathrm{loc}}(\overline{U} \setminus S_\Gamma, \mathcal{N}) \cap C^0(U \setminus (S_\Gamma \cup F), \mathcal{N})$$

with trace h in the local trace sense. For any $x \in \operatorname{Int} K_i$ and any closed transverse disk $\overline{D} \subset U$ centered at x such that $D \cap S_\Gamma = \{x\}$ and $\partial D \cap F = \varnothing$,

$$[v|_{\partial D}]_{\mathcal{N}} = \beta_i.$$

Weighted-length minimality

For $\Gamma = \{(K_i, \beta_i)\}_{i\in I}$, set

$$\mathbb{M}(\Gamma) := \sum_{i\in I} |\beta_i|_* \, \mathcal{H}^1(K_i).$$

Theorem (Homotopical Plateau minimality)

Set

$$\Gamma_* := \{(L_j, \alpha_j) : j \in J\}.$$

Then $\Gamma_* \in \mathcal{C}(\Omega, g)$*, and*

$$\mathbb{M}(\Gamma_*) \le \mathbb{M}(\Gamma) \quad \text{for any } \Gamma \in \mathcal{C}(\Omega, g).$$

The comparison class contains map-realizable labeled networks, not arbitrary unions of segments.

The closest analogue: p-harmonic maps as $p \uparrow 2$

Set $\mathcal{E}_p(v) := \frac{1}{p} \int_\Omega |\nabla v|^p \, \mathrm{d}x$. The two singular perturbations compare as follows.

	p-harmonic limit	present GL limit
Maps	$v_p : \Omega \to \mathcal{N}$	$u_\varepsilon : \Omega \to \mathbb{R}^m$
Scale	$\mathcal{E}_p(v_p) = O\left(\frac{1}{2-p}\right)$	$E_\varepsilon(u_\varepsilon) = O(\lvert \log \varepsilon \rvert)$
Boundary	fixed singular $\mathcal{N}$-valued trace g	ε-dependent, ambient-valued regularization g_ε

Common limit: If $\#\pi_1(\mathcal{N}) < \infty$, both theories give $W^{1,q}$ control for $q < 2$ and convergence away from a weighted one-dimensional network. Under the corresponding convexity and boundary hypotheses, that network is a finite union of straight segments and a homotopical Plateau minimizer.

What does not transfer: The network and Plateau arguments are adapted from the p-harmonic theory, but its $W^{1,q}$ proof does not apply directly. The potential $\varepsilon^{-2} f(u_\varepsilon)$ and the boundary cores require the Fu–Wang–Wang scheme, potential estimates, and boundary partial regularity.

[Bulanyi–Van Schaftingen–Van Vaerenbergh, 2025]

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The two-dimensional line cost

Let $u \in H^1(B_r^2, \mathbb{R}^m)$ and $g := u|_{\partial B_r^2} \in H^1(\partial B_r^2, \mathbb{R}^m)$. Assume $\operatorname{dist}(g, \mathcal{N}) < \delta_{\mathcal{N}}$, and set

$$\sigma := [\Pi_{\mathcal{N}} \circ g]_{\mathcal{N}}.$$

Lower bound

$$E_\varepsilon(u, B_r^2) + CrE_\varepsilon(u, \partial B_r^2) \geq |\sigma|_* \log \frac{r}{\varepsilon} - C.$$

Matching upper bound

If $g \in H^1(\partial B_r^2, \mathcal{N})$, one can choose $\operatorname{tr} u_\varepsilon = g$ with

$$E_\varepsilon(u_\varepsilon, B_r^2) \leq |\sigma|_* \log \frac{r}{\varepsilon} + C \left(r \|\nabla_\top g\|^2_{L^2(\partial B_r^2)} + 1 \right).$$

The coefficient of the logarithmic term is therefore the exact line tension: $|\sigma|_*$.

Interior monotonicity

For an interior point x, set

$$\theta_\varepsilon(u; x, r) := \frac{1}{r} \int_{B_r(x)} e_\varepsilon(u).$$

For Euler–Lagrange solutions, θ_ε is monotone in the interior:

$$\theta_\varepsilon(u; x, t) - \theta_\varepsilon(u; x, s) = \int_s^t \left[\frac{1}{\rho} \int_{\partial B_\rho(x)} |\partial_r u|^2 \, \mathrm{d}\mathcal{H}^2 \right. \\ \left. + \frac{2}{\varepsilon^2 \rho^2} \int_{B_\rho(x)} f(u) \right] \mathrm{d}\rho \geq 0.$$

Passage to the limit. If $\mu_{\varepsilon_i} \overset{*}{\rightharpoonup} \mu_*$, then

$$r \longmapsto \frac{\mu_*(\overline{B}_r(x))}{2r} \quad \text{is non-decreasing,} \qquad \Theta^1(\mu_*, x) := \lim_{r \downarrow 0} \frac{\mu_*(\overline{B}_r(x))}{2r}.$$

Interior clearing-out

Clearing-out estimate

For an interior local minimizer with $\|u\|_{L^\infty(B_{2r}(x))} \leq M$, there exist $\delta \in (0,1)$ and $C = C(f, \mathcal{N}, M) > 0$ such that

$$\varepsilon < \delta r, \quad \theta_\varepsilon(u; x, r) \leq \delta \log \frac{r}{\varepsilon} \quad \Longrightarrow \quad \theta_\varepsilon\left(u; x, \frac{r}{2}\right) \leq C.$$

At the limit, the density gap reads

$$\mu_*(\overline{B}_r(x)) < \delta r \quad \Longrightarrow \quad \mu_*\left(B_{\frac{r}{2}}(x)\right) = 0.$$

Consequently, every $x \in S_* \cap \Omega$ carries uniformly positive density.

Rectifiability and straight segments

Density detection: Clearing-out and monotonicity give

$$S_* \cap \Omega = \{x : \Theta^1(\mu_*, x) \geq \delta_0\}.$$

Stationary varifold:
The limiting stress–energy tensor induces a stationary 1-varifold V in the interior, with approximate unit tangent τ:

$$\delta V(\xi) = \int_{S_* \cap \Omega} (\tau \otimes \tau) : D\xi \,\mathrm{d}\mu_* = 0 \quad \forall\, \xi \in C_c^1(\Omega, \mathbb{R}^3).$$

Structure:

clearing-out + monotonicity + stress–energy compactness

$\implies$ rectifiable stationary 1-varifold,

cylindrical quantization + Allard–Almgren

$\implies$ locally finite straight network.

Density quantization in a cylinder

A thin cylinder $\Lambda_{\delta,1} := B^2_\delta \times (-1,1)$ crosses a single segment (after blow-up). Its lateral trace has class σ, and the excess energy is $\eta \ll 1$. Then

$$E_{\bar\varepsilon}(u_\varepsilon, \Lambda_{\delta,1}) = 2|\sigma|_* \log\frac{\delta}{\bar\varepsilon} + O\left((\delta + \eta + \delta^{-1}\eta)\log\frac{\delta}{\bar\varepsilon}\right) + O(1).$$

- The lower bound follows by slicing into transverse disks.
- The upper bound uses Luckhaus interpolation and an optimal core vortex.
- First let $\bar\varepsilon \downarrow 0$, and then $\eta, \delta \downarrow 0$.

$$\Theta^1(\mu_*, x) = |\sigma|_* \quad \text{for } \mathcal{H}^1\text{-a.e. } x \in S_*.$$

Boundary monotonicity

For $x_0 \in \partial U$, set

$$U_r(x_0) := U \cap B_r(x_0), \quad T_r^U(x_0) := \partial U \cap B_r(x_0), \quad \theta_\varepsilon^U(u; x_0, r) := \frac{1}{r} \int_{U_r(x_0)} e_\varepsilon(u).$$

Let U be bounded and $C^{2,1}$. Assume that u solves the Euler–Lagrange equation in $U_{2r}(x_0)$ and $u = g$ on $T_{2r}^U(x_0)$, where

$$g \in C^1(T_{2r}^U(x_0), \mathcal{N}), \quad r\|\nabla_\top g\|_{L^\infty} \le M_0,$$

and $\theta_\varepsilon^U(u; x_0, 2r) \le M_1$. For $0 < s < t < r$,

Almost-monotonicity

$$\theta_\varepsilon^U(u; x_0, t) - \theta_\varepsilon^U(u; x_0, s) + \alpha(M_0, M_1, r)(t - s) \ge \int_s^t \mathcal{R}_\varepsilon(\rho)\,\mathrm{d}\rho \ge 0,$$

where

$$\mathcal{R}_\varepsilon(\rho) := \frac{1}{\rho} \int_{U \cap \partial B_\rho(x_0)} |\partial_r u|^2 \,\mathrm{d}\mathcal{H}^2 + \frac{2}{\varepsilon^2 \rho^2} \int_{U_\rho(x_0)} f(u),$$

$$\alpha(M_0, M_1, r) := \frac{CM_0}{r} \left(M_0 + M_1^{\frac{1}{2}} \right) + C(M_0^2 + M_1).$$

Boundary clearing-out

Half-ball estimate

Let U be bounded and $C^{2,1}$, and let $g \in H^1(T_{2r}^U(x_0), \mathbb{R}^m)$. Assume that u is a local minimizer in $U_{2r}(x_0)$ up to $T_{2r}^U(x_0)$, with trace g, and

$$\|u\|_{L^\infty(U_{2r})} + \|g\|_{L^\infty(T_{2r}^U)} + E_\varepsilon(g, T_{2r}^U) \le M.$$

Here g may be ambient-valued; no $\mathcal{N}$-valued trace assumption is required. Then there exist $\delta \in (0,1)$ and $C > 0$ such that

$$\varepsilon < \delta r, \quad \theta_\varepsilon^U(u; x_0, r) < \delta \log \frac{r}{\varepsilon} \quad \Longrightarrow \quad \theta_\varepsilon^U\left(u; x_0, \frac{r}{2}\right) \le C.$$

Proof.

- Fubini selects $\rho \in (\frac{r}{2}, r)$ with controlled energy on the spherical side $U \cap \partial B_\rho(x_0)$.
- The cap $T_\rho^U(x_0)$ contributes only $O(1)$ through the boundary energy of g.
- Since $U_\rho(x_0)$ is bi-Lipschitz equivalent to a ball, a Luckhaus extension competitor, together with minimality, gives the bound.

Boundary partial regularity

Regular boundary patch: Away from $\mathcal{A}$, the trace satisfies $g_\varepsilon = g \in \mathcal{N}$ and

$$r\|\nabla_\top g\|_{L^\infty(T^U_{2r})} + r^2\|D^2_\top g\|_{L^\infty(T^U_{2r})} \le M.$$

Assume also $\|u\|_{L^\infty(U_{2r})} + \theta^U_\varepsilon(u; x_0, 2r) \le M$.

Pointwise estimate: For $\varepsilon < \delta r$,

$$(\delta r)^2\|e_\varepsilon(u)\|_{L^\infty(U_{\delta r}(x_0))} \le C, \quad \|f(u)\|_{L^\infty(U_{\delta r}(x_0))} \le C\varepsilon^4 r^{-4}.$$

How the good scale is found:

- Almost-monotonicity and a dyadic pigeonhole argument give a scale s_0 with small density drop and small $s_0\|\nabla_\top g\|_\infty$.
- After flattening, a blow-up limit is a 0-homogeneous minimizing $\mathcal{N}$-valued map on a half-ball with constant trace; hence by [Schoen–Uhlenbeck, 1983] it is constant.
- Boundary ε-regularity then gives the two pointwise bounds.

Why convexity matters

Fix $x_0 \in \partial\Omega \setminus \mathcal{A}$. On a small boundary patch, $g_\varepsilon = g \in \mathcal{N}$ and

$$\frac{1}{|\log \varepsilon|} \int_{T^{\Omega}_{2r}(x_0)} |\nabla_\top g_\varepsilon|^2 \, \mathrm{d}\mathcal{H}^2 \longrightarrow 0.$$

One-sided first variation at the boundary

Set $T^{ij}_\varepsilon(u) := e_\varepsilon(u)\delta_{ij} - \partial_i u \cdot \partial_j u$. If

$$\frac{T_\varepsilon(u_\varepsilon)}{|\log \varepsilon|} \, \mathrm{d}x \overset{*}{\rightharpoonup} T_*,$$

then

$$\int_{\Omega_r(x_0)} \langle D\xi, \mathrm{d}T_* \rangle \geq 0 \quad \text{whenever } \xi \cdot \nu \leq 0 \text{ on } T^{\Omega}_r(x_0).$$

Barrier argument. Boundary clearing-out gives a positive density gap. Together with interior rectifiability and the one-sided variation, strong convexity excludes contact at x_0 [White, 2010]. Hence

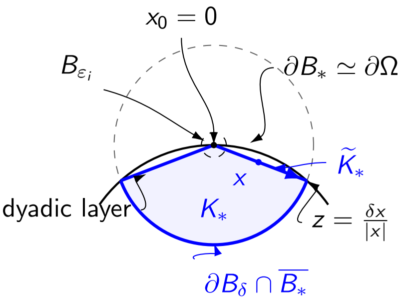
$$S_* \cap (\partial\Omega \setminus \mathcal{A}) = \varnothing.$$

Identifying the boundary endpoints

1. **No spurious boundary points:** The preceding first-variation and barrier arguments give $S_* \cap \partial\Omega \subset \mathcal{A}$.
2. **Every prescribed charge occurs:** If $a_i \notin S_*$, boundary clearing-out and partial regularity produce a regular collar. The non-trivial loop $g|_{\partial D_{a_i}}$ can then be pushed to an interior loop bounding a disk on which u_* is continuous and $\mathcal{N}$-valued—a contradiction.
3. **The network stays in the convex hull:** At a point maximizing the distance from $\mathrm{Conv}(\mathcal{A})$, the force-balance law requires every branch to be tangent to the corresponding level set. Stationarity then rules out such a component.

$$S_* \cap \partial\Omega = \mathcal{A}, \quad S_* \subset \mathrm{Conv}(\mathcal{A}).$$

The regular boundary collar also identifies each boundary charge with the class of an enclosing interior section.



Local model. After a fixed local $C^{2,1}$ change of coordinates, move x_0 to 0 and use

$$B_* := \{y : |y + e_3| < 1\},$$

tangent to $\partial\Omega$. For a good radius δ, set

$$C_\delta := \partial B_\delta \cap \overline{B_*},$$

$$K_* := \left\{ y \in B_* \cap B_\delta : \frac{\delta y}{|y|} \in C_\delta \right\},$$

$$\widetilde{K}_* := \partial K_* \setminus \partial B_\delta.$$

Radial extension

$$v_i(x) := u_{\varepsilon_i}\left(\frac{\delta x}{|x|}\right),$$

$$x \in K_* \setminus B_{\varepsilon_i}.$$

Equality forces radial branches

For radii avoiding branching points, define the total spherical density

$$h(r) := \int_{S_* \cap \partial B_r} \Theta^1(\mu_*, x)\, \mathrm{d}\mathcal{H}^0(x).$$

Cone comparison: The dyadic layer is negligible at logarithmic scale. At a good radius δ, comparison with the radial extension gives

$$\int_0^\delta h(r)\, \mathrm{d}r \le \delta h(\delta).$$

A minimizing radius: Choose δ where h attains its minimum. Since $h(r) \ge h(\delta)$ on $(0, \delta)$, the reverse inequality holds. Consequently,

$$\frac{\mu_*(\Omega_r)}{r} = h(\delta) \qquad (0 < r < \delta).$$

Equality in the monotonicity formula gives $(x - x_0)^\perp = 0$ μ_*-a.e.; hence only finitely many radial straight branches meet x_0.

Two regularity scales

Let $U \subset \mathbb{R}^3$ and $x \in U$. Two scales encode different notions of regularity.

Pointwise regularity scale

$$r(u;x) := \sup\left\{0 \le s \le 1 : s^2 \|e_\varepsilon(u)\|_{L^\infty(U_s(x))} \le 1\right\}.$$

Set $\mathrm{Bad}_{\mathrm{I}}(u;r) := \{x : r(u;x) < r\}$.

Energy regularity scale For a fixed threshold $\Lambda > 0$, define

$$r^\Lambda(u;x) := \sup\left\{0 \le s \le 1 : \theta_\varepsilon^U(u;x,s) \le \Lambda\right\},$$

and

$$\mathrm{Bad}_{\mathrm{II}}(u;r,\Lambda) := \{x : r^\Lambda(u;x) < r\}.$$

Clearing-out converts control of $\mathrm{Bad}_{\mathrm{II}}$ into a volume estimate for $\mathrm{Bad}_{\mathrm{I}}$.

Three geometric regions

Boundary case: Once $S_* \cap \partial\Omega = \mathcal{A}$ is known, choose finitely many boundary patches, each containing at most one charge $x_0 \in \mathcal{A}$. At scale r, cover

$$\mathrm{Bad}_{\mathrm{II}}(u_\varepsilon; r, \Lambda) \cup \{x_0\}$$

by at most Cr^{-1} balls of radius r.
The domain then splits into three regions:

1. **Near the regular boundary**: The trace is smooth and $\mathcal{N}$-valued, so boundary partial regularity gives a uniform scale.
2. **In the interior:** Use the interior covering and interior regularity estimates. We actually use arguments in [Fu–Wang–W., 2025, 2026].
3. **Inside the covering balls:** Their total volume is $Cr^{-1}r^3 = Cr^2$.

Distribution estimate: Fix $q \in (1,2)$ and choose $\gamma \in (\frac{q}{2}, 1)$. Then

$$\mathcal{L}^3\big(\{x : r(u_\varepsilon; x) < \delta R\}\big) \le CR^{2\gamma}, \quad \nabla u_\varepsilon \in L^{2\gamma,\infty} \hookrightarrow L^q \quad (q < 2\gamma).$$

Controlling the potential energy

Pointwise decay. Partial regularity gives

$$f(u_\varepsilon) \le C\varepsilon^4 r^{-4}.$$

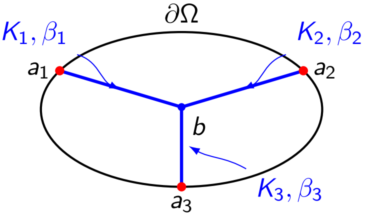
Split the integral into

- the regular region, using pointwise decay;
- the bad region, using the covering-volume bound;
- boundary collars, using partial regularity.

For an $O(\varepsilon)$ core around a one-dimensional network, multiscale summation gives

$$\int_\Omega f(u_\varepsilon)\,\mathrm{d}x \le C\varepsilon^2.$$

Fix an arbitrary $\Gamma = \{(K_i, \beta_i)\} \in \mathcal{C}(\Omega, g)$ with a realizing map v.



Recovery sequence

$$w_i = g_{\varepsilon_i} \quad \text{on } \partial\Omega,$$

$$\limsup_{i\to\infty} \frac{E_{\varepsilon_i}(w_i, \Omega)}{|\log \varepsilon_i|} \leq \mathbb{M}(\Gamma).$$

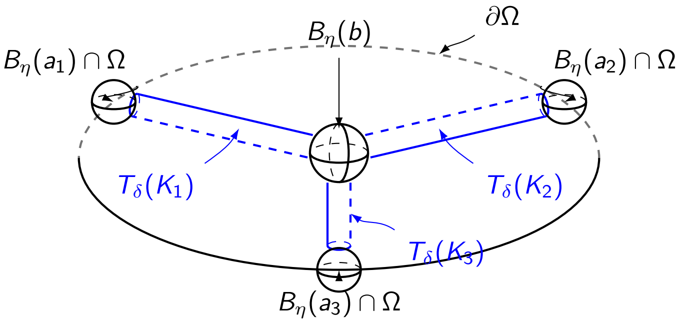
Global minimality: Since u_{ε_i} is a global minimizer,

$$E_{\varepsilon_i}(u_{\varepsilon_i}) \leq E_{\varepsilon_i}(w_i).$$

Since $\mu_*(\overline{\Omega}) = \mathbb{M}(\Gamma_*)$, passage to the limit gives

$$\mathbb{M}(\Gamma_*) \leq \mathbb{M}(\Gamma).$$

Recovery near cylinders and balls



- **Cylinders**: insert optimal class-β_i vortex profiles.
- **Balls**: glue the tube caps, match g_{ε_i}, and retain v outside.

$$\limsup_{i\to\infty} \frac{E_{\varepsilon_i}(w_i, \Omega)}{|\log \varepsilon_i|} \leq \sum_i |\beta_i|_* \mathcal{H}^1(K_i) = \mathbb{M}(\Gamma).$$

Outline

1. Background and setting
2. Topological energy and network cost
3. Main results
4. Key estimates and proof strategy
5. Summary

Two scales in the proof

Compactness scale:

$E_\varepsilon \lesssim |\log \varepsilon| \implies$ monotonicity and clearing-out $\implies$ rectifiable stationary limit.

Topological scale:

two-dimensional line cost $\implies$ density quantization $\implies$ finite weighted network.

Minimality and uniform estimates:

recovery in cylinders and balls $\implies \mathbb{M}(\Gamma_*) \leq \mathbb{M}(\Gamma)$,

regular scales and multiscale covers $\implies$ $W^{1,q}$ and potential bounds.

Conclusion

For three-dimensional global minimizers with finite $\pi_1(\mathcal{N})$, the normalized energy converges to

$$\mu_* = \sum_{j\in J} |\alpha_j|_* \, \mathcal{H}^1 \llcorner L_j, \qquad S_* = \bigcup_{j\in J} L_j.$$

The network is finite, its densities are quantized, and

$$S_* \cap \partial\Omega = \mathcal{A}, \qquad S_* \subset \mathrm{Conv}(\mathcal{A}).$$

Uniformly for every $q < 2$,

$$\|\nabla u_\varepsilon\|_{L^q(\Omega)} \le C_q, \qquad \frac{1}{\varepsilon^2}\int_\Omega f(u_\varepsilon)\,\mathrm{d}x \le C.$$

Finally, Γ_* solves the homotopical Plateau problem

$$\mathbb{M}(\Gamma_*) \le \mathbb{M}(\Gamma) \quad \text{for every } \Gamma \in \mathcal{C}(\Omega, g).$$

Topology fixes the admissible charges; global minimality selects the geometry.

Thank you for your attention!